United States Patent Application Publication 20250277581
Kind Code A1
Publication Date September 04, 2025
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LED LIGHT SOURCE DEVICE AND BIOCHEMICAL ANALYSIS DEVICE

Abstract

A biochemical analysis device that emits light used to measure absorbance of a sample includes an LED that emits the light, a substrate on which the LED is mounted, a spacer that supports the substrate, a holder that accommodates the LED, the substrate, and the spacer, and a flow path formed by circulating constant temperature water through the holder. The spacer is formed of a material having a heat conductivity higher than that of the holder. The spacer is in contact with the holder at a contact portion. A convex portion of the spacer is covered with a concave portion of the holder through an air layer.

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Family ID: **88835182**

Appl. No.: **18/857879**

Filed (or PCT Filed): **March 15, 2023**

PCT No.: **PCT/JP2023/009998**

Foreign Application Priority Data

JP	2022-080025	May. 16, 2022

Publication Classification

Int. Cl.: **F21V29/56** (20150101); **F21V29/503** (20150101); **G01N21/31** (20060101)

U.S. Cl.:

CPC **F21V29/56** (20150115); **F21V29/503** (20150115); **G01N21/31** (20130101); G01N2201/062 (20130101)

Background/Summary

TECHNICAL FIELD
[0001] The present invention relates to an LED light source device and a biochemical analysis device.
BACKGROUND ART
[0002] PTL 1 discloses a biochemical analysis device using an LED light source instead of using a tungsten lamp having a short lamp life as a light source. PTL 1 discloses that “The housing is a mechanism that supports the LED substrate and the light pipe. The housing may be made of plastic or metal. Preferably, when formed of a metal having high heat conductivity, for example, aluminum, it can be used as a path for dissipating heat of the LED substrate.”. This literature also describes that “A heat sink may be provided because the LED module, particularly the LED element, generates heat around it. In addition, a cooling mechanism using water cooling or air cooling may be provided on the opposite side of the heat sink to the LED module.”. Further, it is also described that “The efficiency with which the phosphor absorbs light having a near-ultraviolet wavelength to a blue light wavelength and emits light tends to decrease when the temperature of the phosphor increases. Therefore, as described above, the light source is desirably provided with a cooling mechanism.”.
CITATION LIST
Patent Literature
[0003] PTL 1: JP 2020-87974 A
SUMMARY OF INVENTION
Technical Problem
[0004] The LED module of the biochemical analysis device described in PTL 1 includes two components, namely a housing formed of a metal having high heat conductivity such as aluminum, and a cooling mechanism using water cooling or air cooling on the opposite side of the heat sink to the LED module. Having the two components makes it possible to shorten the time required to stabilize the light amount of the LED light source after the power is turned on and improve the analysis throughput. On the other hand, when the LED module is installed in an environment where temperature fluctuation is large, there is a problem that the temperature of the LED module also fluctuates and the luminance efficiency of the LED finally fluctuates.
[0005] An object of the present invention is to provide a biochemical analysis device that achieves both the requirement to shorten the time required to stabilize the light amount of an LED light source after the power is turned on and the requirement to suppress the temperature fluctuation of an LED module installed under an environment where temperature fluctuation is large to suppress fluctuation of luminance efficiency of an LED.
Solution to Problem
[0006] In order to solve the above problems, one of the representative biochemical analysis devices according to the present invention is achieved by including an LED that emits the light, a substrate on which the LED is mounted, a spacer that supports the substrate, a housing (hereinafter, holder) that accommodates the LED, the substrate, and the spacer, and a flow path formed by circulating water through the holder, wherein the spacer is formed of a material having a heat conductivity higher than that of the holder, the spacer is in contact with the holder at a contact portion, and a convex portion of the spacer is covered with a concave portion of the holder through an air layer.
[0007] In order to solve the above problems, one of the representative LED light source devices

according to the present invention is an LED light source device that emits light used to measure absorbance of a sample and includes an LED that emits the light, a substrate on which the LED is mounted, a spacer that supports the substrate, and a holder that accommodates the LED, the substrate, and the spacer, wherein the spacer is formed of a material having a heat conductivity higher than that of the holder, the spacer is in contact with the holder at a contact portion, and a convex portion of the spacer is covered with a concave portion of the holder through an air layer.
Advantageous Effects of Invention
[0008] According to the present invention, since the spacer is formed of a material having a low heat conductivity and the spacer and the holder are in contact with each other, the time required for the light amount of the LED light source to be stabilized can be shortened.
[0009] In addition, by covering the convex portion of the spacer with the concave portion of the holder, the exposed area of the spacer to the outside air is reduced, and by forming the air layer between the concave portion of the holder and the convex portion of the spacer, the thermal resistance is increased, and the temperature fluctuation of the LED module installed in an environment where the temperature fluctuation is large can be suppressed, thereby suppressing the fluctuation of the luminance efficiency of the LED.
[0010] Problems, configurations, and effects other than those described above will be clarified by the following description of embodiments.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0011] FIG. **1** is a view showing the overall configuration of a biochemical analysis device to which the present invention is applied.
[0012] FIG. **2** is a sectional view for explaining the configuration of an LED unit according to the first embodiment.
[0013] FIG. **3** is a sectional view of a holder and a spacer for explaining a heat dissipation path according to a first embodiment.
[0014] FIG. **4** is a schematic view for explaining convective heat transfer of a cylindrical object.
[0015] FIG. **5**A is a sectional view for explaining the configuration of a spacer according to the first embodiment.
[0016] FIG. **5**B is a sectional view for explaining the configuration of the holder according to the first embodiment.
[0017] FIG. **5**C is a sectional view for explaining the configuration of the spacer and a sub-base according to the first embodiment.
[0018] FIG. **6** is a schematic diagram for explaining the heat conduction of a cylindrical object.
[0019] FIG. **7** is a sectional view for explaining thermal resistance related to the heat conduction of the spacer according to the first embodiment.
[0020] FIG. **8** is a diagram for explaining a breakdown of thermal resistance related to the heat conduction of the spacer according to the first embodiment.
[0021] FIG. **9**A is a side view for explaining thermal resistance related to the heat conduction of the flat plate portion of the spacer according to the first embodiment.
[0022] FIG. **9**B is a sectional view for explaining thermal resistance related to the heat conduction of the flat plate portion of the spacer according to the first embodiment.
[0023] FIG. **10**A is a side view for explaining thermal resistance related to the heat conduction of the inner cylindrical portion and the outer cylindrical portion of the spacer according to the first embodiment.
[0024] FIG. **10**B is a sectional view for explaining thermal resistance related to the heat conduction of the inner cylindrical portion and the outer cylindrical portion of the spacer according to the first

embodiment.

[0025] FIG. **11**A is a graph showing fluctuations in ambient temperature according to the first embodiment.

[0026] FIG. **11**B is a graph showing fluctuations in constant temperature water temperature according to the first embodiment.

[0027] FIG. **12** is a graph illustrating substrate temperatures when constant temperature water having the ambient temperature and constant temperature water having the water temperature illustrated in FIGS. **11**A and **11**B are injected into holders.

[0028] FIG. **13** is a diagram illustrating standard deviations of substrate temperatures for 10 minutes when constant temperature water having the ambient temperature and constant temperature water having the water temperature illustrated in FIGS. **11**A and **11**B are injected into the holders.

DESCRIPTION OF EMBODIMENTS

[0029] Hereinafter, embodiments of the present invention will be described with reference to the accompanying drawings.

First Embodiment

[0030] Hereinafter, an embodiment of the present invention will be described with reference to FIGS. **1** and **2**. First, the overall configuration of a biochemical analysis device **100** according to the present embodiment will be described with reference to FIG. **1**. FIG. **1** is a diagram showing the overall configuration of the biochemical analysis device **100** according to the first embodiment.

<Configuration of Biochemical Analysis Device>

[0031] The biochemical analysis device **100** includes a conveyance line **101**, a rotor **102**, a reagent disk **103**, a reaction disk **104**, a pipetting device **105**, a mixing device **106**, a spectrometer **107**, a reaction cell washing device **108**, a nozzle washing device **109**, a control unit **115**, an input unit **123**, a display unit **124**, and the like.

[0032] The conveyance line **101** transfers a specimen rack **111** holding a specimen container **110** containing a specimen to a specimen pipetting position **121** by a required amount. The pipetting device **105** pipettes the specimen from the specimen container **110** to the reaction cell **112** (reaction container) at the specimen pipetting position **121**. The conveyance line **101** is further connected to a rotor **102**. By rotating the rotor **102**, the specimen rack **111** is exchanged with another conveyance line **101**.

[0033] The reagent disk **103** holds a reagent bottle **113** containing a reagent and rotationally transfers the reagent bottle **113** to a position where the pipetting device **105** can perform a pipetting operation. The pipetting device **105** pipettes a reagent from the reagent bottle **113** to the reaction cell **112** at a reagent pipetting position **122**. The reagent is pipetted into the reaction cell **112** in an amount necessary for colorimetric analysis and reacts with component in a specimen to be analyzed.

[0034] The reaction disk **104** holds the reaction cell **112** and rotationally transfers the reaction cell **112**, which is a target of each operation, to operation positions indicating positions where the spectrometer **107** that performs colorimetric analysis, the mixing device **106**, the reaction cell washing device **108**, and the like respectively operate. The reaction cell **112** is kept warm by a constant temperature medium such as water. As a result, in the reaction liquid that is a mixture of the specimen and the reagent, the chemical reaction between the component in the specimen and the reagent is promoted.

[0035] The pipetting device **105** sucks a specimen to be subjected to colorimetric analysis from the specimen container **110** and discharges the specimen to the reaction cell **112**. The pipetting device **105** sucks a reagent corresponding to an analysis target from the reagent bottle **113** and discharges the reagent to the reaction cell **112**. The pipetting device **105** includes an arm **118**, a nozzle **116**, and a pipetting device motor **119**. The arm **118** holds the nozzle **116** and the liquid level sensor **117**. The nozzle **116** is connected to a liquid level sensor **117**. The liquid level sensor **117** detects the presence or absence of liquid by a capacitance change. A shield portion **114** is installed near the

position where the pipetting device **105** performs a pipetting operation. The pipetting device motor **119** moves the pipetting device **105** in the vertical direction or the rotational direction.
[0036] The mixing device **106** mixes the reaction liquid in the reaction cell **112** in order to promote the reaction between the component to be analyzed in the specimen discharged from the specimen container **110** to the reaction cell **112** and the reagent discharged from the reagent bottle **113** to the reaction cell **112**.
[0037] An LED unit **120** irradiates the reaction liquid mixed and chemically reacted by the mixing device **106** with light. The spectrometer **107** disperses the transmitted light that has passed through the reaction liquid. Colorimetric analysis by absorbance measurement is performed based on the dispersed transmitted light.
[0038] The reaction cell washing device **108** sucks the reaction liquid from the reaction cell **112** for which the colorimetric analysis has been completed, discharges detergent or the like, and washes the reaction cell **112**.
[0039] The nozzle washing device **109** washes the tip of the nozzle **116** of the pipetting device **105** into which the specimen or the reagent has been pipetted. As a result, the residue adhering to the nozzle **116** is removed, and the next analysis target is not affected. The control unit **115** includes a processor, a memory, and the like and controls each device and the like.
[0040] The input unit **123** includes a keyboard, a mouse, a touch panel, and the like and inputs an instruction from the user to the control unit **115**. The display unit **124** includes a liquid crystal display (LCD) or the like and displays an operation screen or the like.
<Configuration of LED Unit>
[0041] A peripheral structure of the LED unit **120** according to the present embodiment will be described with reference to FIG. **2**. FIG. **2** is a sectional view for explaining the configuration of the LED unit **120** according to the embodiment.
[0042] The LED unit **120** has a structure in which a member such as a substrate **402** is accommodated in a holder **401**. The light emitted from the LED unit **120** is output from an optical path **404**. Thus, the optical path **404** has an opening at least on the optical axis (i.e. at a position along the arrow next to reference numeral **404** in FIG. **2**).
[0043] An LED **405** is mounted on the substrate **402**. A spacer **406** is a member that supports the substrate **402** and is in mechanical contact with the holder **401**. By adjusting the position of the spacer **406**, the optical axis of the LED **405** can be aligned with the opening of the optical path **404** and the optical axis of an optical component **407**. That is, the spacer **406** serves as a member that supports the substrate **402** and aligns the optical axis.
[0044] The light emitted from the LED **405** is multiplexed by a multiplexing device **408** and emitted toward the optical path **404** via the optical component **407** (for example, a lens or a slit). This light is output from the opening of the optical path **404** via the optical component **407** (for example, a lens or a slit).
[0045] A water channel **411** for circulating constant temperature water for adjusting the temperature of the LED unit **120** is formed inside the wall of the holder **401**. This constant temperature water enters from an inlet **409** and exits from an outlet **410**. The temperature of the constant temperature water is set to, for example, a temperature higher than the outside air temperature of the LED unit **120**. In this case, if the temperature is desired to be lowered, the amount of heat to be propagated is reduced by reducing the amount of water, and the temperature is naturally lowered. Therefore, it is not necessary to separately provide an element or the like capable of cooling operation, and there is an advantage that the temperature can be adjusted with a simple configuration. In addition, the constant temperature liquid may be, for example, a constant temperature liquid used in a constant temperature bath provided for the biochemical analysis device to keep the temperature of the sample constant. In this case, by supplying the constant temperature liquid to the LED unit **120**, the LED unit **120** need not include its own constant temperature liquid supply source, which is useful. In this case, the inlet **409** and the outlet **410** are connected to the constant temperature bath. When

the constant temperature liquid is shared with the automatic analysis device in this manner, control of the flow rate of constant temperature liquid and the like may be entrusted to the automatic analysis device. Alternatively, for example, the LED unit **120** itself may have a flow rate control function by arranging a flow rate regulating valve or the like inside the LED unit **120**. A device such as a controller that gives a control instruction may be incorporated inside the LED unit **120** or may be configured as a device outside the LED unit **120**.

[0046] The optical path **404** has an opening through which light emitted from the LED **405** passes. This opening allows air to enter and exit between the internal space of the LED unit **120** and the outside air. Such entrance and exit of air becomes a factor that fluctuates the temperature of the internal space of the LED unit **120**, and thus is desirably suppressed as much as possible. That is, it can be said that it is desirable to keep the opening size of the optical path **404** to the minimum necessary for emitting light.

[0047] In addition to the opening of the optical path **404**, a hole capable of ventilating between the internal space of the LED unit **120** and the outside air may be provided. For example, a hole through which wiring for supplying power to the LED **405** passes may be provided in a part of the spacer **406**. Even in this case, the opening size of the hole is desirably smaller than the opening size of the optical path **404**. That is, it is desirable that the ventilation capacity of the internal space of the LED unit **120** is maximized by the opening size of the optical path **404**. As a result, the heat exchange between the internal space and the outside air is suppressed to the minimum, so that the temperature fluctuation of the LED **405** can be suppressed. Even in a case where a ventilable hole is provided other than the opening of the optical path **404**, if the hole can be sealed with an appropriate sealing material or the like, it is desirable to seal the hole.

<LED Operation>

[0048] When the power is applied to the LED unit **120**, the LED **405** emits light, and the temperature of the LED **405** and its peripheral members gradually rises with the lapse of time. When the temperature reaches a certain level, the temperature is stabilized.

[0049] The light amount and the wavelength of the LED **405** have temperature characteristics and fluctuate depending on the temperature. Therefore, in order to quickly stabilize the characteristics of the emitted light, it is necessary to quickly raise the temperatures of the LED and its peripheral members, particularly the spacer **406** and the substrate **402**. For this purpose, it is necessary to reduce the volume of the spacer **406** as much as possible to reduce the heat capacity as much as possible and to configure the substrate **402** and the spacer **406** with a material (for example, aluminum) such as a metal having high heat conductivity.

<Heat Dissipation of LED>

[0050] Next, a heat dissipation path when the LED is turned on will be described with reference to FIG. **3**. As illustrated in FIG. **3**, the heat generated when a current flows through the LED **405** is transferred from the LED **405** to the substrate **402** and from the substrate **402** to the spacer **406**. There are two subsequent heat transfer paths, the first being a heat dissipation path from a surface **418** of the spacer which is exposed to the outside air to the outside air (heat dissipation path (**1**) in FIG. **3**), and the second being a heat dissipation path from the spacer **406** to the holder **401** and from the holder **401** to the constant temperature water in the water channel **411** (heat dissipation path (**2**) in FIG. **3**).

<Problem of Heat Dissipation Path (**1**)>

[0051] Hereinafter, the heat dissipation path (**1**) will be described. In the heat dissipation path (**1**), heat is dissipated from the surface **418** of the spacer which is exposed to the outside air to the outside air. The airflow of the outside air may fluctuate due to disturbance. Due to the fluctuation in the airflow, the amount of heat dissipated from the surface **418** of the spacer which is exposed to the outside air to the outside air fluctuates, the temperature of the spacer **406** fluctuates. Furthermore, the temperatures of the substrate **402** and the LED **405** fluctuate. Finally, the luminance efficiency of the LED **405** fluctuates. In order to prevent the occurrence of the temperature fluctuation of the LED even in the case of such disturbance, it is effective that the spacer **406** and the holder **401** are formed of a member having a large heat capacity or a small heat conductivity.

[0052] On the other hand, as described above, in order to quickly raise the temperature of the LED **405** to a stable state, the spacer **406** needs to be formed of a member having a small heat capacity and a large heat conductivity. That is, the measure for the heat capacity and the heat conductivity of the spacer **406** and the holder **401** to reduce the temperature fluctuation of the LED even when there is a disturbance is opposite to the measure for rapidly increasing the temperature of the LED to the stable state.

[0053] Therefore, in order to achieve both the requirement to reduce the temperature fluctuation of the LED and the requirement to quickly raise the temperature of the LED to the stable state even when there is a disturbance, in the present embodiment, (a) the thermal resistance of convective heat transfer is increased, and (b) the thermal resistance of heat conduction is adjusted, with the spacer **406** being a member having a large heat conductivity. Hereinafter, (a) and (b) will be described in detail.

< (a) Increase in Thermal Resistance of Convective Heat Transfer>

[0054] First, (a) increasing the thermal resistance of convective heat transfer will be described. FIG. **4** is a schematic diagram of convective heat transfer in cylindrical object. In convective heat transfer, the relationship among a surface temperature Ts, a fluid temperature Tg, a heat flow rate Qt, and a thermal resistance Rt is expressed by (Equation 1).

[00001] $(Ts - Tg) = \mathrm{Rt} \times \mathrm{Qt}$ (Equation1)

[0055] Here, the thermal resistance Rt is expressed by (Equation 2) from the convective heat transfer coefficient hm and a contact area Aout with the outside air.

[00002] $\mathrm{Rt} = 1/(\mathrm{hm} \times \mathrm{Aout})$ (Equation2)

[0056] In order to increase the thermal resistance Rt, it is found from (Equation 2) that the contact area Aout with the outside air may be reduced.

[0057] Here, in the present invention, in order to reduce the contact area Aout with the outside air, the spacer **406** forms a convex portion **412** as illustrated in FIG. **5**A, and the holder **401** forms a concave portion **413** as illustrated in FIG. **5**B. Further, as illustrated in FIG. **5**C, the outer periphery of the spacer convex portion **412** is covered with the holder concave portion **413**. With such a configuration, it is possible to reduce the contact area of the spacer **406** with the outside air and to increase the thermal resistance of convective heat transfer. As a result, even when there is a disturbance, the temperature fluctuation of the spacer **406** can be reduced, and furthermore, the temperature fluctuation of the substrate **402** and the LED **405** can also be reduced. As a result, it is possible to achieve both the requirement to rapidly increase the temperature of the LED **405** and the requirement to reduce the temperature fluctuation even when there is a disturbance.

[0058] Further, by forming the water channel **411** in the holder concave portion **413**, the water channel **411** can be brought close to the spacer convex portion **412** (shown in FIG. **5**C). As a result, the contribution of the heat dissipation path (**2**) (heat dissipation from the spacer **406** to the holder **401** and from the holder **401** to the constant temperature water in the water channel **411**) becomes larger than that of the heat dissipation path (**1**) (heat dissipation from the surface **418** of the spacer which is exposed to the outside air to the outside air), and the temperature fluctuation can be reduced even in a case where there is disturbance.

< (b) Adjustment of Thermal Resistance of Heat Conduction>

[0059] Next, (b) adjustment of thermal resistance of heat conduction will be described. FIG. **6** is a schematic diagram of the heat conduction of a cylindrical object. In heat conduction, the relationship among a temperature difference (T**1**−T**2**), a heat flow rate Pc, and a thermal resistance Rc is expressed by (Equation 3).

[00003] $(T1 - T2) = \mathrm{Rc} \times \mathrm{Pc}$ (Equation3)

[0060] Here, the thermal resistance Rc is expressed by (Equation 4) from the length L, the heat conductivity λ, and the sectional area A.

[00004] $Rc = L / (\quad R)$ (Equation4)

[0061] In order to increase the thermal resistance Rc, it is found from (Equation 4) that the length L may be increased, or the heat conductivity λ or the sectional area A may be decreased.

[0062] Here, with reference to FIGS. **7**, **8**, **9**A, and **9**B, thermal resistance related to the heat conduction of the spacer **406** according to the present embodiment will be described.

[0063] FIG. **7** is a sectional view of the holder **401** and the spacer **406** for explaining thermal resistance related to the heat conduction of the spacer in the present embodiment. As illustrated in FIG. **7**, a thermal resistance Rsp related to heat conduction in the present embodiment is from the LED **405** to the surface **418** of the spacer which is exposed to the outside air (the gray area in FIG. **7**).

[0064] As illustrated in FIG. **8**, the thermal resistance Rsp is divided into a thermal resistance Rfl (the dot portion in FIG. **8**) from the LED **405** to an inner cylindrical portion **415** in a flat plate portion **414**, a thermal resistance Rin (the hatched portion in FIG. **8**) of the inner cylindrical portion **415**, and a thermal resistance Rout (the grid portion in FIG. **8**) of an outer cylindrical portion **416** and is calculated from (Equation 5).

[00005] $Rsp = Rfl + Rin + Rout$ (Equation5)

[0065] Here, the flat plate portion **414** is a portion that supports the substrate **402** on which the LED **405** is mounted. The inner cylindrical portion **415** is a portion for positioning the spacer **406** with respect to the holder **401** by being fitted into a cylindrical portion of the holder **401** to align the optical axis. Further, the outer cylindrical portion **416** is a portion that comes into contact with the holder **401** and transfers heat between the spacer **406** and the holder **401**.

[0066] The thermal resistance Rfl will be described below with reference to FIGS. **9**A and **9**B. Assuming that the length from the LED **405** to the inner cylindrical portion **415** in the flat plate portion **414** is Lfl (FIG. **9**A), the sectional area of the flat plate portion **414** is Afl (the hatched area in FIG. **9**B), and the heat conductivity of the flat plate portion **414** is kfl, the thermal resistance Rfl is expressed by (Equation 4) to (Equation 6).

[00006] $Rfl = Lfl / (Afl \times kfl)$ (Equation6)

[0067] Similarly, the thermal resistance Rin of the inner cylindrical portion **415** and the thermal resistance Rout of the outer cylindrical portion **416** will be described with reference to FIGS. **10**A and **10**B. When the length of the inner cylindrical portion **415** is represented by Lin, the length of the outer cylindrical portion **416** is represented by Lout (both shown in FIG. **10**A), the diameter of the inner cylindrical portion **415** is represented by din, the diameter of the outer cylindrical portion **416** is represented by dout (both shown in FIG. **10**B), the heat conductivity of the inner cylindrical portion **415** is represented by kin, and the heat conductivity of the outer cylindrical portion **416** is represented by kout, the thermal resistances Rin and Rout are expressed by (Equation 7) and (Equation 8), respectively.

[00007] $Rin = Lin / [(din / 2)^2 \times \quad \times kin]$ (Equation7)

$Rout = Lout / [\{(dout / 2)^2 - (din / 2)^2\} \times \quad \times kout]$ (Equation8)

[0068] By using (Equation 5) to (Equation 8) given above, the thermal resistance Rsp from the LED **405** to the surface **418** of the spacer which is exposed to the outside air is obtained.

[0069] Here, as indicated by (Equation 6), (Equation 7), and (Equation 8), in order to increase the thermal resistances Rfl, Rin, and Rout, the lengths Lfl, Lin, and Lout may be increased, or the sectional area Afl, the diameters din and dout, and the heat conductivities kfl, kin, and kout may be decreased.

[0070] Specifically, the thermal resistance Rsp is preferably 2 (K/W) or more and 40 (K/W) or less in order to achieve both the requirement to reduce the temperature fluctuation of the LED even

when there is a disturbance and the requirement to rapidly raise the temperature of the LED to the stable state.
[0071] In addition, the spacer **406** may be formed of two or more kinds of materials, and the heat conductivity of any one of kfl, kin, and kout may be reduced to increase the thermal resistance Rsp.
[0072] Further, only the heat conductivity kout of the outer cylindrical portion **416** having the surface **418** of the spacer which is exposed to the outside air may be reduced to suppress the heat dissipation to the outside air, or the outer cylindrical portion **416** may be formed of two or more kinds of materials to reduce the heat conductivity of the portion having the surface **418** of the spacer which is exposed to the outside air. Examples of a material having a low heat conductivity include brass, bronze, stainless steel, and PEEK. By forming a film on the surface **418** of the spacer which is exposed to the outside air, the heat conductivity of only the surface **418** of the spacer which is exposed to the outside air may be reduced to suppress heat dissipation to the outside air.
Effects
[0073] Effects of the present embodiment will be described with reference to FIGS. **11**A, **11**B, **12**, and **13**. FIGS. **11**A and **11**B are graphs illustrating temporal changes in the ambient temperature of the LED unit and the water temperature, FIG. **12** is a graph illustrating temporal changes in the LED substrate temperature, and FIG. **13** is a graph illustrating the standard deviation of the LED substrate temperature for 10 minutes, respectively. The 10-minute standard deviation of the LED substrate temperature, which is an index illustrated in FIG. **13**, is an index for determining whether the LED substrate temperature has rapidly increased after the power of the LED is turned on. It is known that the standard deviation of the LED substrate temperature for 10 minutes decreases with time after the LED power is turned on, and when the standard deviation decreases to about 0.01° C., the light amount and the wavelength of the LED are also stabilized.
[0074] The ambient temperature is assumed to fluctuate by 2° C. as shown in FIG. **11**A, and the water temperature is assumed to fluctuate by about 0.1° C. as shown in FIG. **11**B. In the conventional configuration in which the thermal resistance Rsp is 40 or more so as to reduce the influence of the LED substrate temperature under the temperature environment illustrated in FIGS. **11**A and **11**B, the LED substrate temperature can be suppressed to a fluctuation of about 0.09° C. as illustrated in FIG. **12**, but the time required for the standard deviation of the LED substrate temperature for 10 minutes to be 0.01° C. or less becomes about 980 seconds as illustrated in FIG. **13**.
[0075] On the other hand, in the present invention in which the thermal resistance Rsp is about 3, the fluctuation in the LED substrate temperature can be suppressed to about 0.11° C. as shown in FIG. **12**, and the time required for the standard deviation of the LED substrate temperature for 10 minutes to be 0.01° C. or less can be set to about 125 seconds as shown in FIG. **13**.
REFERENCE SIGNS LIST
[0076] **100** biochemical analysis device [0077] **101** conveyance line [0078] **102** rotor [0079] **103** reagent disk [0080] **104** reaction disk [0081] **105** pipetting device [0082] **106** mixing device [0083] **107** spectrometer [0084] **108** reaction cell washing device [0085] **109** nozzle washing device [0086] **110** specimen container [0087] **111** specimen rack [0088] **112** reaction cell [0089] **113** reagent bottle [0090] **114** shield portion [0091] **115** control unit [0092] **116** nozzle [0093] **117** liquid level sensor [0094] **118** arm [0095] **119** pipetting device motor [0096] **120** LED unit [0097] **121** specimen pipetting position [0098] **122** reagent pipetting position [0099] **123** input unit [0100] **124** display unit [0101] **201** input/storage unit [0102] **202** ISE unit [0103] **203** specimen conveyance unit [0104] **204** colorimetric analysis unit [0105] **303** reaction tank [0106] **304** diffraction grating [0107] **305** photodetector [0108] **306** absorbance calculator [0109] **307** current detection unit [0110] **308** current adjustment unit [0111] **401** holder [0112] **402** substrate [0113] **404** optical path [0114] **405** LED [0115] **406** spacer [0116] **407** optical component [0117] **408** multiplexing device [0118] **409** inlet [0119] **410** outlet [0120] **411** water channel [0121] **412** spacer convex portion [0122] **413** holder concave portion [0123] **414** flat plate portion [0124] **415** inner

cylindrical portion [0125] **416** outer cylindrical portion [0126] **417** contact surface [0127] **418** surface of spacer which is exposed to outside air

Claims

1. A biochemical analysis device that emits light used to measure absorbance of a sample, the biochemical analysis device comprising: an LED that emits the light; a substrate on which the LED is mounted; a spacer that supports the substrate; a holder that accommodates the LED, the substrate, and the spacer; and a flow path formed by circulating constant temperature water through the holder, wherein the spacer is formed of a material having a heat conductivity higher than that of the holder, the spacer is in contact with the holder at a contact portion, and a convex portion of the spacer is covered with a concave portion of the holder through an air layer.
2. The biochemical analysis device according to claim 1, wherein when a thermal resistance from an LED installation portion of the spacer to a contact portion of the spacer with the outside air is Rsp, a relationship of 2 (K/W)≤Rsp≤40 (K/W) is satisfied.
3. The biochemical analysis device according to claim 1, wherein the spacer is formed of not less than two materials.
4. The biochemical analysis device according to claim 3, wherein a material of the spacer is different between a contact portion with outside air and other portions, and a heat conductivity of a material of the contact portion with the outside air is smaller than a heat conductivity of other portions.
5. The biochemical analysis device according to claim 1, wherein a film is formed at a contact portion of the spacer with outside air.
6. The biochemical analysis device according to claim 5, wherein the heat conductivity of the film is smaller than the heat conductivity of the spacer.
7. The biochemical analysis device according to claim 1, wherein a flow path is disposed in the concave portion of the holder.
8. The biochemical analysis device according to claim 1, wherein the holder has an internal temperature higher than an outside air temperature.
9. The biochemical analysis device according to claim 1, wherein a flow rate of constant temperature water flowing through a water channel is reduced, so that a temperature inside the holder is lowered.
10. An LED light source device that emits light used to measure absorbance of a sample, the LED light source device comprising: an LED that emits the light; a substrate on which the LED is mounted; a spacer that supports the substrate; and a holder that accommodates the LED, the substrate, and the spacer, wherein the spacer is formed of a material having a heat conductivity higher than that of the holder, the spacer is in contact with the holder at a contact portion, and a convex portion of the spacer is covered with a concave portion of the holder through an air layer.
11. The LED light source device according to claim 10, wherein when a thermal resistance from an LED installation portion of the spacer to a contact portion of the spacer with the outside air is Rsp, a relationship of 2 (K/W)≤Rsp≤40 (K/W) is satisfied.
12. The LED light source device according to claim 10, wherein the spacer is formed of not less than two materials.
13. The LED light source device according to claim 10, wherein a material of the spacer is different between a contact portion with outside air and other portions, and a heat conductivity of a material of the contact portion with the outside air is smaller than a heat conductivity of other portions.
14. The LED light source device according to claim 10, wherein a film is formed at a contact portion of the spacer with outside air.

15. The LED light source device according to claim 14, wherein the heat conductivity of the film is smaller than the heat conductivity of the spacer.